

Clean before you count

Analysis done on dirty data is not analysis. It is decoration.

Every dataset arrives dirty. The question is never *whether* yours has problems, but whether you find them **before** or **after** you publish. This is a checklist you run once, systematically, before you calculate a single statistic.

The seven checks

1. Count your rows

Does the number of records match the number you expect from your register? If your register says 180 and your file has 181, you have a duplicate.

Excel: =COUNTA(A2:A200) • R: nrow(df)

2. Hunt for duplicates

The same patient entered twice inflates every count you produce.

Excel: Data > Remove Duplicates (on a COPY!) • R: sum(duplicated(df\$record_id))

3. Check every category list

'M', 'Male', 'male' and '1' are four spellings of one category. Excel counts them as four groups. Sort the column, or pivot it, and read every distinct value.

Excel: PivotTable with the field in Rows • R: table(df\$sex, useNA = 'ifany')

4. Check minimum and maximum

An age of 199 or a negative number of days is an entry error. The mean may barely move, which is exactly why you must check the extremes and not rely on the average.

Excel: =MIN(range) and =MAX(range) • R: summary(df)

5. Check the logic between columns

Does any treatment end date fall before its start date? Is anyone pregnant and male? These are impossible combinations and each one is a real error.

Excel: =IF(P2<O2,"CHECK","") • R: which(df\$end < df\$start)

6. Count the blanks, column by column

Missing data is not a nuisance — it is a result. Know exactly how much you have and where it is, because it changes every denominator.

Excel: =COUNTBLANK(range) • R: colSums(is.na(df))

7. Check that numbers are stored as numbers

One cell containing '12 km' turns the whole column into text in some operations, and AVERAGE() will skip it silently. Numbers align right in Excel; text aligns left.

Excel: =COUNT(range) vs =COUNTA(range) • R: str(df)

The two rules of cleaning

Rule 1 — Never overwrite your raw data

Keep the original file untouched and read-only. Do all cleaning in a **copy**, or better, in a script. If you clean by hand and make a mistake, you can never go back and you will never know what you changed.

Rule 2 — Write down every change you make

Keep a cleaning log: what you changed, which records, and why. When a reviewer asks 'how did you handle the 5 patients with missing weight?' you need an answer, and 'I cannot remember' is not one.

Cleaning log — copy this table into your own analysis file

Date	Column	What I found	What I did	Why

Liafuan importante • Key words in Tetum

Clean (data)	Hamoos dadus
Mistake, error	Sala
Check, verify	Verifika
Duplicate, doubled	Duplika
Empty, blank	Mamuk
Change	Muda